

Calculus 3

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May 8, 2024

Outline

- 1 Surface Integrals of type 1
- 2 Surface Integrals of type 2
- 3 Gauss-Ostrogradski theorem
- 4 Stokes theorem

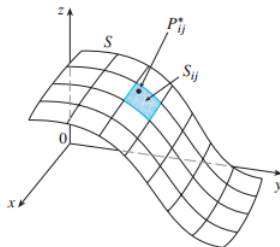
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Surface Integrals of type 1

Calculate the volume of the curved surface

Given a surface S in $\mathbb{R}^3$ with an unequal mass distribution according to its surface area is described as a mass function per unit area $f(x, y, z)$. Calculate the mass M of the surface S .



The surface S is divided into corresponding patches S_{ij} . We evaluate f at a point P_{ij}^* in each patch, multiply by the area ΔS_{ij} of the patch, and form the Riemann sum

$$M \approx \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*, z_{ij}^*) \Delta S_{ij}.$$

Surface Integrals of type 1

Then, we take the limit as the number of patches increases and define the **surface integral of type 1 of f over the surface S** as

$$\iint_S f(x, y, z) dS = \lim_{m, n \rightarrow \infty} \sum_{i=1}^m \sum_{j=1}^n f(x_{ij}^*, y_{ij}^*, z_{ij}^*) \Delta S_{ij}.$$

The properties of Surface Integrals of type 1

1. $\iint_S \alpha f(x, y, z) dS = \alpha \iint_S f(x, y, z) dS$, where $\alpha \in \mathbb{R}$.
2. $\iint_S \alpha [f(x, y, z) + g(x, y, z)] dS = \iint_S \alpha f(x, y, z) dS + \iint_S \alpha g(x, y, z) dS$.
3. If $S = S_1 \cup S_2$ then

$$\iint_S f(x, y, z) dS = \iint_{S_1} f(x, y, z) dS + \iint_{S_2} f(x, y, z) dS.$$

Remark: The surface integral of type 1 does not depend on the side of the surface.

How to compute the surface integral of type 1

Suppose that a surface S has a vector equation

$$\mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}, \quad (u, v) \in D.$$

Then, we define

$$\mathbf{r}_u = \frac{\partial x}{\partial u}\mathbf{i} + \frac{\partial y}{\partial u}\mathbf{j} + \frac{\partial z}{\partial u}\mathbf{k}, \quad \mathbf{r}_v = \frac{\partial x}{\partial v}\mathbf{i} + \frac{\partial y}{\partial v}\mathbf{j} + \frac{\partial z}{\partial v}\mathbf{k}.$$

Compute the surface integral of type 1

If the components are continuous and $\mathbf{r}_u$ and $\mathbf{r}_v$ are nonzero and nonparallel in the interior of D then we have

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA.$$

How to compute the surface integral of type 1

Example: Compute the surface integral $\iint_S x^2 dS$, where S is the unit sphere $x^2 + y^2 + z^2 = 1$.

How to compute the surface integral of type 1

If the surface S is given by

$$z = z(x, y)$$

and D_{xy} is the projection of S onto the plane Oxy . Then,

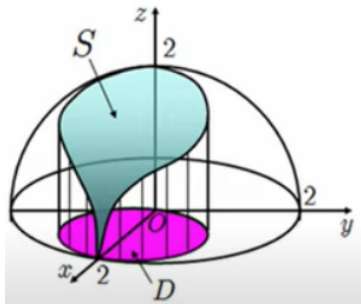
$$dS = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy$$

and

$$\iint_S f(x, y, z) dS = \iint_{D_{xy}} f(x, y, z(x, y)) \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy.$$

How to compute the surface integral of type 1

Example: Evaluate $\iint_S z \, dS$, where S is part of a half sphere $x^2 + y^2 + z^2 = 4, z \geq 0$ and lies within a circular cylinder $x^2 + y^2 - 2x = 0$.



How to compute the surface integral of type 1

If the surface S is given by

$$y = y(x, z)$$

and D_{xz} is the projection of S onto the plane Oxz . Then,

$$dS = \sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2} dx dz$$

and

$$\iint_S f(x, y, z) dS = \iint_{D_{xz}} f(x, y(x, z), z) \sqrt{1 + \left(\frac{\partial y}{\partial x}\right)^2 + \left(\frac{\partial y}{\partial z}\right)^2} dx dz.$$

How to compute the surface integral of type 1

If the surface S is given by

$$x = x(y, z)$$

and D_{yz} is the projection of S onto the plane Oyz . Then,

$$dS = \sqrt{1 + \left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2} dy dz$$

and

$$\iint_S f(x, y, z) dS = \iint_{D_{xy}} f(x(y, z), y, z) \sqrt{1 + \left(\frac{\partial x}{\partial y}\right)^2 + \left(\frac{\partial x}{\partial z}\right)^2} dy dz.$$

How to compute the surface integral of type 1

Example: Evaluate $I = \iint_S \frac{1}{(1+x+y)^2} dS$, where S is the surrounding face of the tetrahedron $x + y + z = 1, x = 0, y = 0, z = 0$.

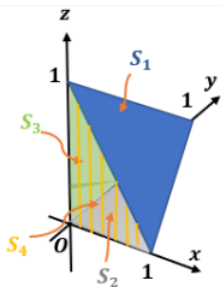


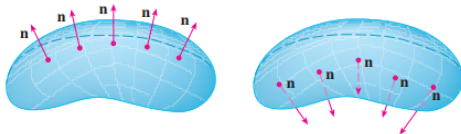
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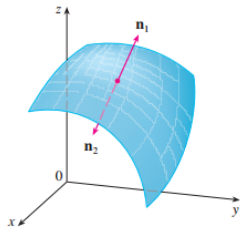
Surface Integrals of type 2

The oriented surface

If it is possible to choose a unit normal vector $\vec{n}$ at every such point (x, y, z) so that $\vec{n}$ varies continuously over S , then S is called an **oriented surface** and the given choice of $\vec{n}$ provides S with an **orientation**. There are two possible orientations for any orientable surface



Remark: There are two unit normal vectors $\vec{n}_1$ and $\vec{n}_2 = -\vec{n}_1$ at (x, y, z)



Surface Integrals of type 2

Let S is given by $F(x, y, z) = 0$. Then, the unit normal vector $\vec{n}$ of S is

$$\vec{n} = \left(\frac{F'_x}{\sqrt{(F'_x)^2 + (F'_y)^2 + (F'_z)^2}}, \frac{F'_y}{\sqrt{(F'_x)^2 + (F'_y)^2 + (F'_z)^2}}, \frac{F'_z}{\sqrt{(F'_x)^2 + (F'_y)^2 + (F'_z)^2}} \right)$$

In the special case, when the surface S is determined by the equation $z = z(x, y)$ and

- The unit normal vector $\vec{n}$ is **directed upwards**, that is, the unit normal vector creates an acute angle with the Oz -axis and the third coordinate of $\vec{n}$ is positive. Then, the surface S is given by $F(x, y, z) = z - z(x, y) = 0$ and

$$\vec{n} = \left(\frac{-z'_x}{\sqrt{1 + (z'_x)^2 + (z'_y)^2}}, \frac{-z'_y}{\sqrt{1 + (z'_x)^2 + (z'_y)^2}}, \frac{1}{\sqrt{1 + (z'_x)^2 + (z'_y)^2}} \right)$$

- The unit normal vector $\vec{n}$ is **directed downwards**, that is, the unit normal vector creates an obtuse angle with the Oz -axis and the third coordinate of $\vec{n}$ is negative. Then, the surface S is given by $F(x, y, z) = z(x, y) - z = 0$ and

$$\vec{n} = \left(\frac{z'_x}{\sqrt{1 + (z'_x)^2 + (z'_y)^2}}, \frac{z'_y}{\sqrt{1 + (z'_x)^2 + (z'_y)^2}}, \frac{-1}{\sqrt{1 + (z'_x)^2 + (z'_y)^2}} \right)$$

Surface Integrals of type 2

Definition

The surface S is **smooth** if $F(x, y, z)$ has first-order partial derivatives F'_x, F'_y, F'_z continuous and is not simultaneously 0 on S .

Definition

Let $P(x, y, z), Q(x, y, z), R(x, y, z)$ is defined on the oriented smooth surface S . The unit normal vector $\vec{n}$ of S is $\vec{n} = (\cos(\alpha), \cos(\beta), \cos(\gamma))$ with α, β, γ are the angle formed by $\vec{n}$ and Ox, Oy, Oz -axis, respectively. Then, the surface integrals of type 1

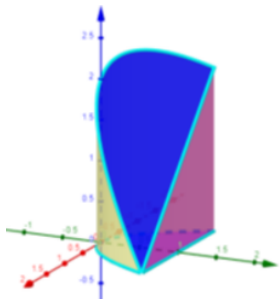
$$I = \iint_S [P(x, y, z) \cos(\alpha) + Q(x, y, z) \cos(\beta) + R(x, y, z) \cos(\gamma)] dS \quad (1)$$

is called **the surface integrals of type 2** for P, Q, R on the oriented surface S . The integral (1) can be rewritten as

$$I = \iint_S P dy dx + Q dz dx + R dx dy.$$

Surface Integrals of type 2

Example: Evaluate $I = \iint_S (x - y) dy dz + z dx dy$, where S is the part of the plane $x + y + z = 2$ that is intercepted by the planes $y = x^2$, $y = 1$, taken above in the direction of the Oz axis.



Surface Integrals of type 2

Example: Evaluate $I = \iint_S zy^2 dy dz + (y + y^2) dz dx + x^2 dx dy$, where S is the part of the paraboloid surface $z = x^2 + y^2$ that lies below the plane $z = 1$, taken below in the direction of the Oz axis.

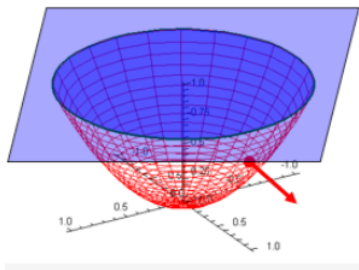


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Gauss-Ostrogradski theorem

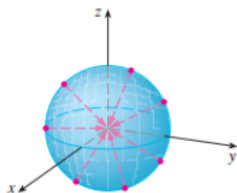
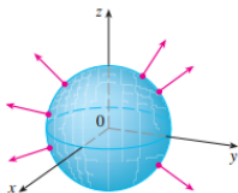
Gauss-Ostrogradski theorem

Let S be the closed surface and Ω be surrounded by S . If

$P(x, y, z)$, $Q(x, y, z)$, $R(x, y, z)$ and its first partial partial derivatives are continuous on Ω , then

$$\iint_S P dy dz + Q dz dx + R dx dy = \pm \iiint_{\Omega} \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} \right) dx dy dz$$

- The sign “+” if the direction of the unit normal vector $\vec{n}$ of the surface S is directed to the outside of Ω .
- The sign “-” if the direction of the unit normal vector $\vec{n}$ of the surface S is directed to the inside of Ω .



Gauss-Ostrogradski theorem

Example: Let $\Omega : x^2 + y^2 \leq z \leq 1$ and the surface S is directed to the outside of Ω . Evaluate

$$I = \iint_S zy^2 dy dz + (y + y^2) dz dx + x^2 dx dy$$

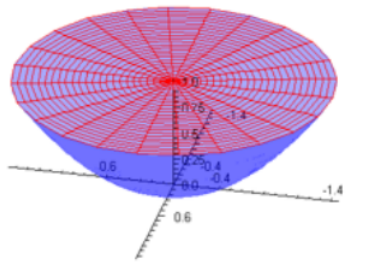


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Stokes theorem

Stokes theorem

Let S be an oriented smooth surface that is bounded by a simple, closed, smooth boundary curve C . Let $P(x, y, z)$, $Q(x, y, z)$, $R(x, y, z)$ and its first partial derivatives are continuous on S . Then,

$$\begin{aligned} & \int_C P(x, y, z) dx + Q(x, y, z) dy + R(x, y, z) dz \\ &= \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy. \end{aligned}$$

Remark: The direction of the unit normal vector $\vec{n}$ of the surface S and the direction of the closed curve C obey the right-hand rule:

- If the unit normal vector $\vec{n}$ of S is **directed upwards**, then the direction of the closed curve C is the counterclockwise direction.
- If the unit normal vector $\vec{n}$ of S is **directed downwards**, then the direction of the closed curve C is the clockwise direction.

Stokes theorem

Example: Use Stokes theorem to compute the integral $I = \int_C 2ydx - xdy + xdz$, where C is the intersection of $x^2 + y^2 = 1$ with the surface $z = y + 1$, taken counterclockwise from the positive side of the Oz axis.

THANK YOU FOR YOUR
ATTENTION !